

ENV 6128

Smart Air Quality Monitoring & Air Pollution Control

Today's Topics

- Visibility
- Expectation
 - Understand different measures of visibility; the major causes of visibility issues and the processes affecting visible radiation; and light extinctions

Measures of Visibility

Introduction

- Visibility: A measure of how far a person can see and distinguish objects
 - horizontal view
- Visibility reduction is an important regional effect of air pollution
 - Aesthetic effect
 - Affect atmospheric chemistry, climate

Introduction



Smog in Shenyang, Liaoning, China. Oct 28th, 2013
<http://news.163.com/photoview/00AP0001/39497.html>

Introduction



LA smog on Dec 19, 2000 (Figure 7.25, Jacobson 2002)

Introduction



Impaired visibility in Shenandoah Valley

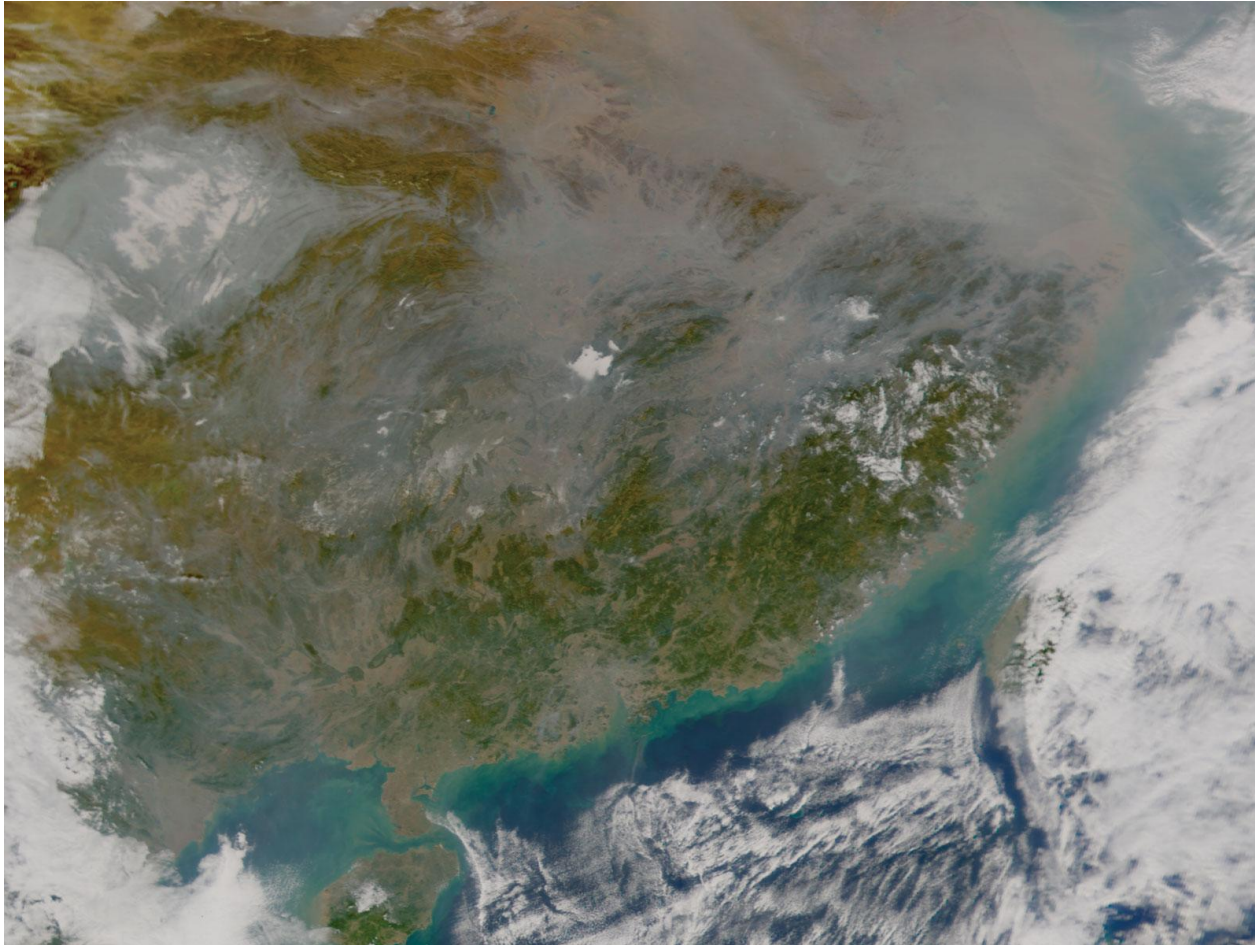
<http://www.nps.gov/media/photo/gallery.htm?id=21C21642-155D-451F-67E688AC226602D3>

Introduction



Arctic haze
(www.awi-bremerhaven.de)

Introduction



Asian brown cloud

<http://kids.britannica.com/elementary/art-124377/A-toxic-mix-of-ash-acids-and-airborne-particles-forming>

Causes of visibility issue

- Light extinction caused by:
 - Scattering
 - Redirection, no attenuation
 - Absorption
 - Some attenuation, energy conversion
- Both gas and particles
 - Both natural and anthropogenic

Total Extinction

- Total Extinction:
 - The attenuation of incident light over a given distance due to absorption and scattering
- Total extinction coefficient
 - Sum of absorption and scattering by gases and particles

$$\sigma_t = \sigma_{a,g} + \sigma_{s,g} + \sigma_{a,p} + \sigma_{s,p}$$

$$C = e^{-\sigma x}$$

Measures of visibility

- Visibility is how far we can see
- Three metrics
 - Meteorological range
 - Prevailing Range
 - Visual Range

Meteorological range

- Meteorological range is a function of Contrast Ratio

$$C = \frac{I_b - I}{I_b}$$

Background
light intensity

Perfect black body.
Absorb all light

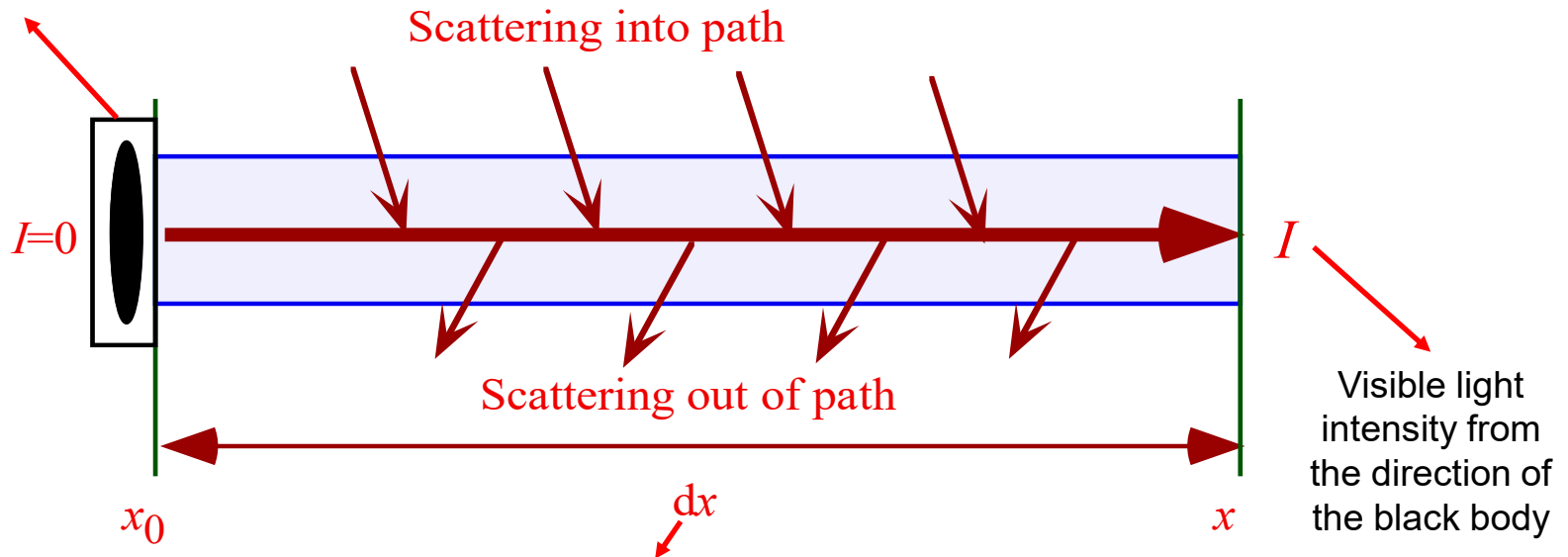


Figure 7.22 Jacobson, 2012

With increase in distance, the blackbody appears
“whiter” due to scattering of light

Contrast Ratio

- At $x = 0$, $I = 0$, contrast ratio = 1. The blackbody is perfectly distinguishable
- Increase in distance $\rightarrow I$ begin to increase $\rightarrow C$ started to decrease
- Threshold contrast – lowest contrast at which people can see $\sim 2\%$
 - Differ by people. Some have better visual acuity
- When $I = I_b$, $C = 0$. The blackbody can not be distinguished

$$C = \frac{I_b - I}{I_b}$$

Meteorological range

- Meteorological range and contrast ratio

$$C = e^{-\sigma x}$$

σ : Extinction coefficient; x : Meteorological range

- Recall threshold contrast for most people: 0.02

$$x = \frac{3.912}{\sigma}$$

- Typical meteorological range:
 - Clean air: 300-400 km. Polluted air: ~ 4 km
- What is the equation for x for someone with threshold contrast of 0.01?

Prevailing Range & Visual Range

- Prevailing Range (V): how far you can see along at least half of your horizon circle

- Empirically:

$$V \sim \frac{1.9}{\sigma}$$

- Visual Range: How far you can see through the air
- Both prevailing range & visual range are subjective
- Now, how to obtain the extinction coefficient?
 - Need to understand what causes visibility issue

Causes of Visibility

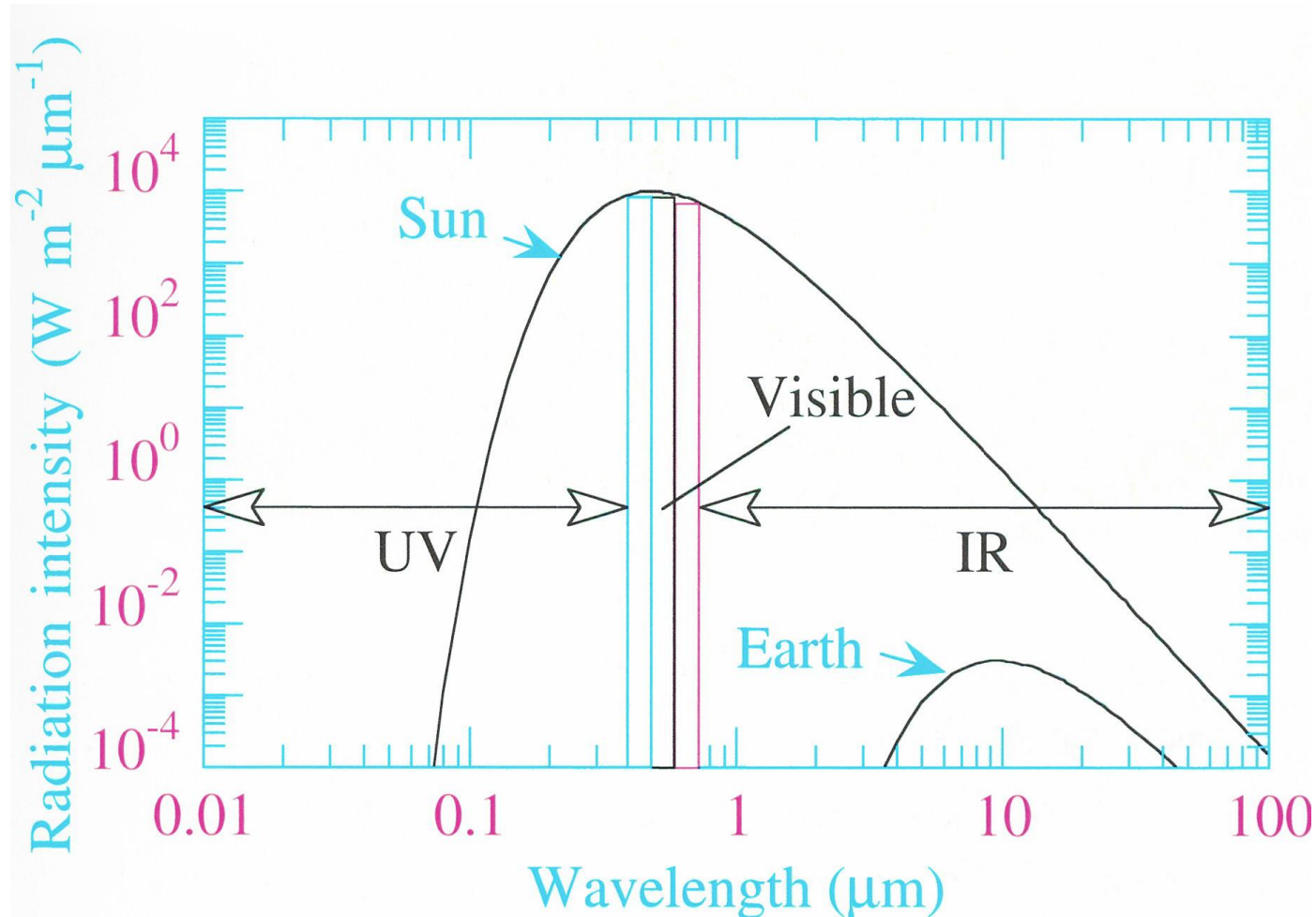
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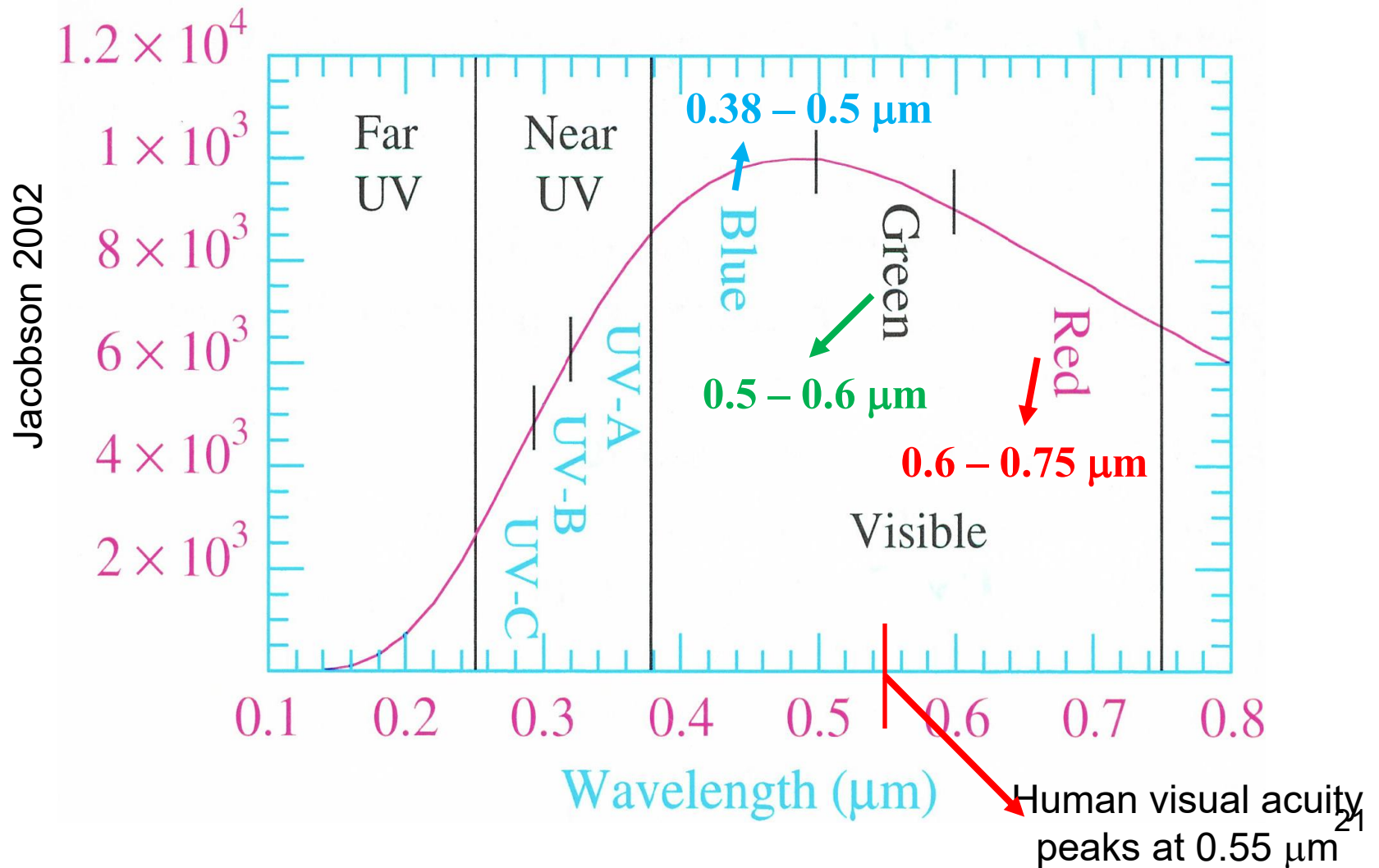
Causes of visibility issue

- In clean air, visibility limited to about 300 ~ 400 kilometers due to air molecule (scattering)
- Visibility depends on altitude due to density (number concentration) of air molecules
- RH has significant effects as hygroscopic particles take up water

The Solar Spectrum



UV and Visible Solar Spectrum



Gas Scattering

- Gas Scattering
 - ▣ Rayleigh Scattering ($D/\lambda < 0.03$)
 - gas molecules $\sim 0.00005 \mu\text{m}$, visible light $\sim 0.5 \mu\text{m}$
 - ▣ All gas scatters
 - N_2 , O_2 , Ar, H_2O most important
 - ▣ Prefer short wavelength (toward blue $\rightarrow$ blue sky)

Gas Absorption

□ Gas Absorption

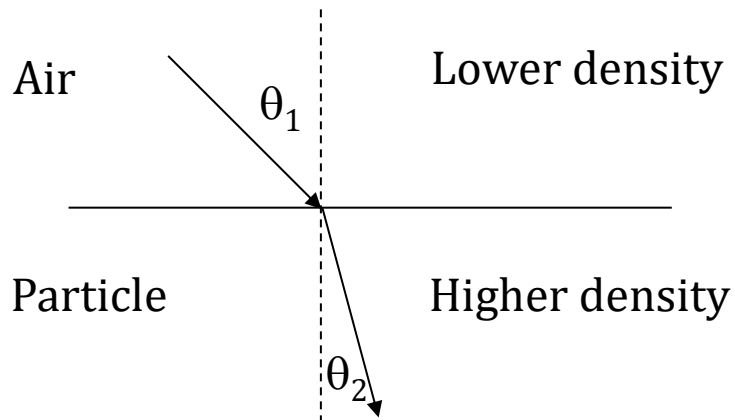
- Only a few gases absorb visible wavelength
- NO_2 ($< 0.71 \mu\text{m}$)
 - Most important gas absorber, especially in polluted air
- NO_3 ($< 0.67 \mu\text{m}$)
 - Quickly go away with presence of sunlight
 - Not very important to visibility
- O_3 ($< 0.35 \text{ UV}, 0.45\text{-}0.75 \mu\text{m}$)
 - Not very important to visibility since relatively weak absorber for visible light

Particle Scattering

- All particles scatter
- Angle of scattering depends on composition, size, and shape of particle
- Scattering is a combination of
 - Refraction
 - Reflection
 - Diffraction

Particle Scattering

Refraction



$$\text{Snell's Law: } n_2 \sin \theta_2 = n_1 \sin \theta_1$$

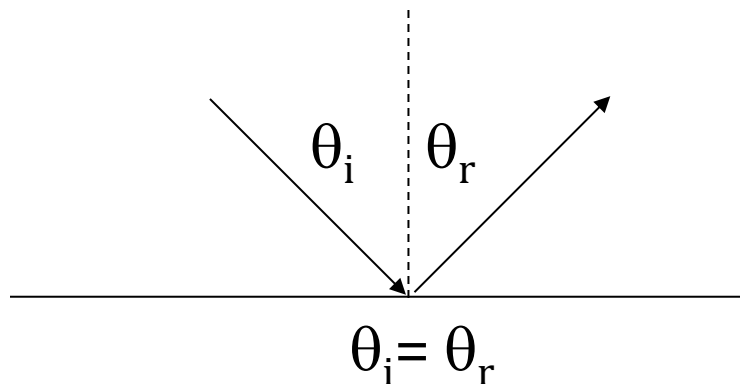
n = real index of refraction

$$= c_{\text{vacuum}} / c_{\text{medium}}$$

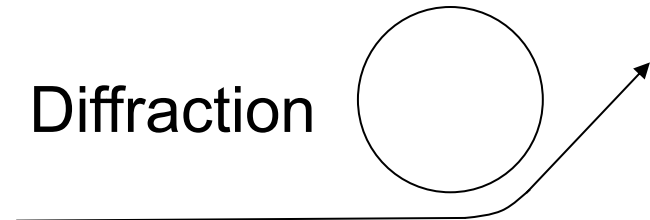
c = speed of light

$$= 3.0 \times 10^8 \text{ m/s in vacuum}$$

Reflection



Diffraction



Particle Scattering

A: reflected

B: refracted

C: diffracted

D: refracted, reflected twice, then refracted

E: refracted, reflected once, then refracted

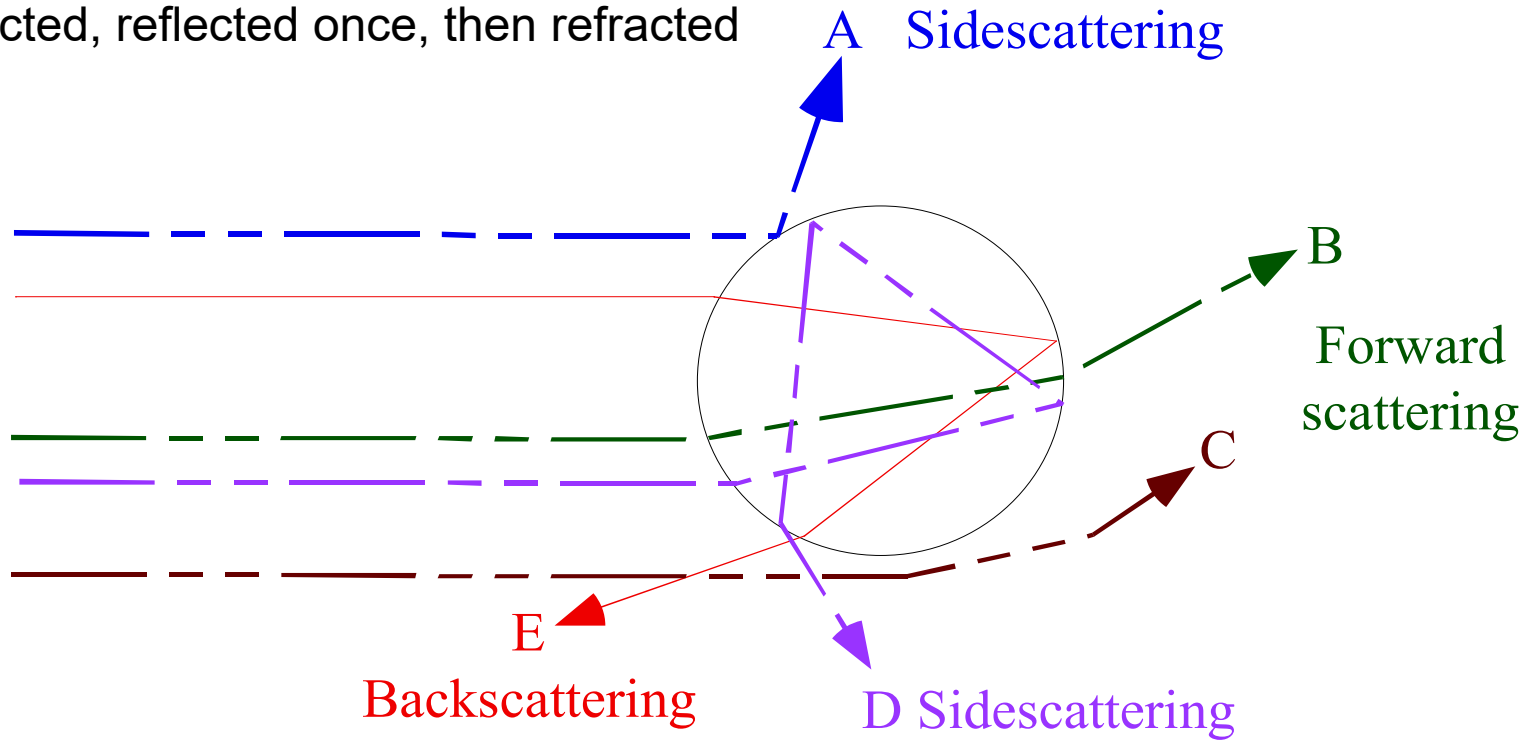


Figure 7.16, Jacobson 2012

Index of refraction

Table 16.1 Index of Refraction of Some Common Materials at $\lambda = 0.589 \mu\text{m}$, the Wavelength of Yellow Sodium Light^{a)}.

Material	Index of Refraction	Material	Index of Refraction
Vacuum	1.0	Quartz (SiO_2)	1.544
Water vapor	1.00025	Polystyrene latex	1.590
Air	1.00028	Diamond	2.417
Water	1.3330	Urban aerosol (avg.)	$1.56 - 0.087i$
Glycerine	1.4730	Methylene blue	$1.55 - 0.6i$
Benzene	1.501	Soot	$1.96 - 0.66i$
Ice (H_2O)	1.305	Carbon ^{b)}	$2.0 - 10i$
Glass	1.52–1.88	Iron	$2.80 - 3.34i$
Sodium chloride	1.544	Copper	$0.47 - 2.81i$

^{a)} Data taken primarily from *Handbook of Chemistry and Physics*, 100th ed., CRC, Boca Raton, FL (2019).

^{b)} $\lambda = 0.491 \mu\text{m}$.

Particle Scattering

- Real index of refraction varies for different wavelengths of light
- Particle in different size ranges scatter differently
 - Smaller than wavelength ($D/\lambda < 0.03$): Rayleigh scattering (forward and backward). Generally, for particles $< 0.05 \mu\text{m}$
 - Same order as wavelength, $0.03 < D/\lambda < 32$: Mie scattering (preferentially forward)
 - More forward as particle size increases
 - Generally particle $> 100 \mu\text{m}$ geometric scattering

Particle Scattering

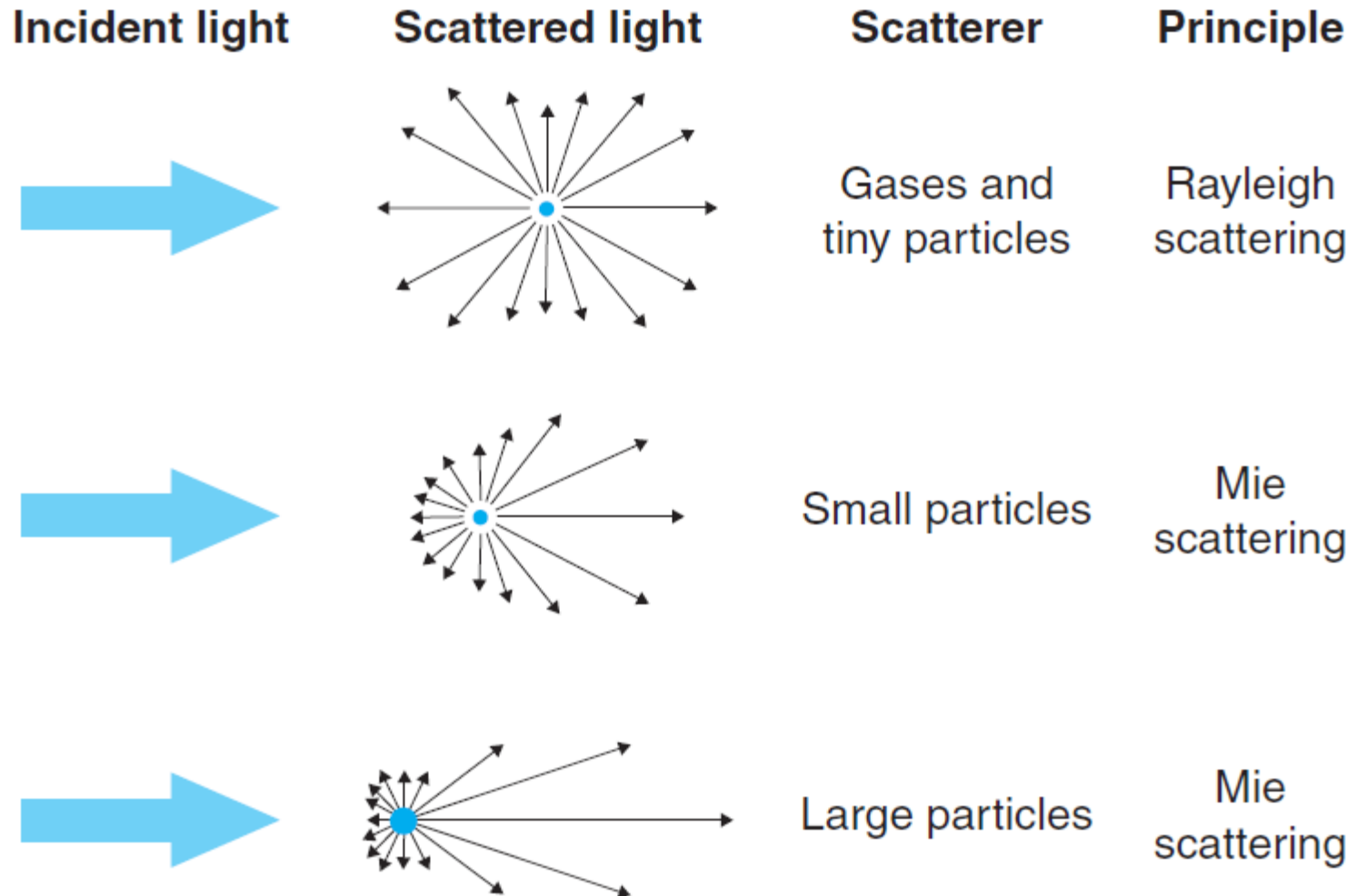


Figure 5-1 Phalen

Rayleigh Scattering

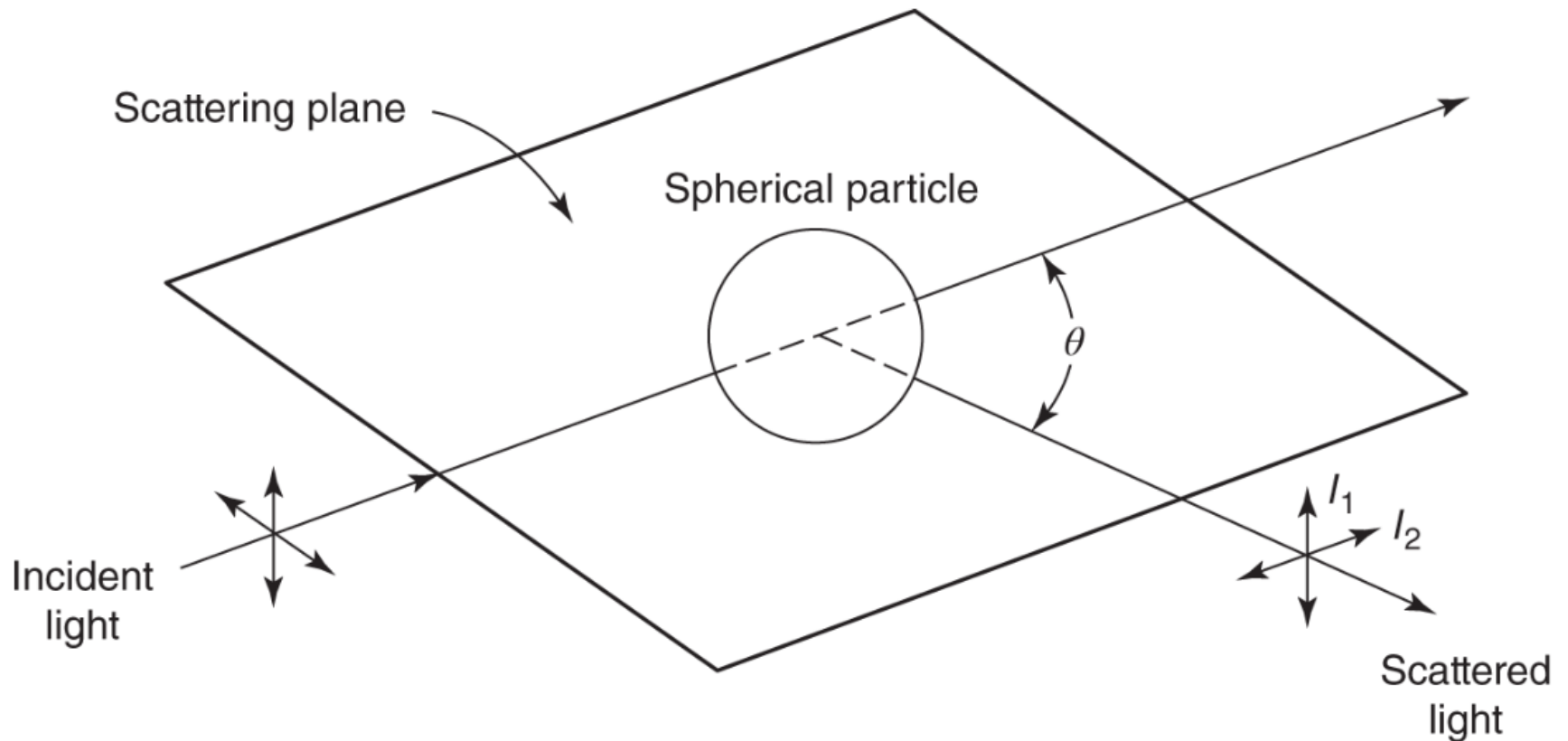


Figure 16.6 Diagram showing scattering angle, scattering plane, and the polarized components of scattered light.

Rayleigh Scattering

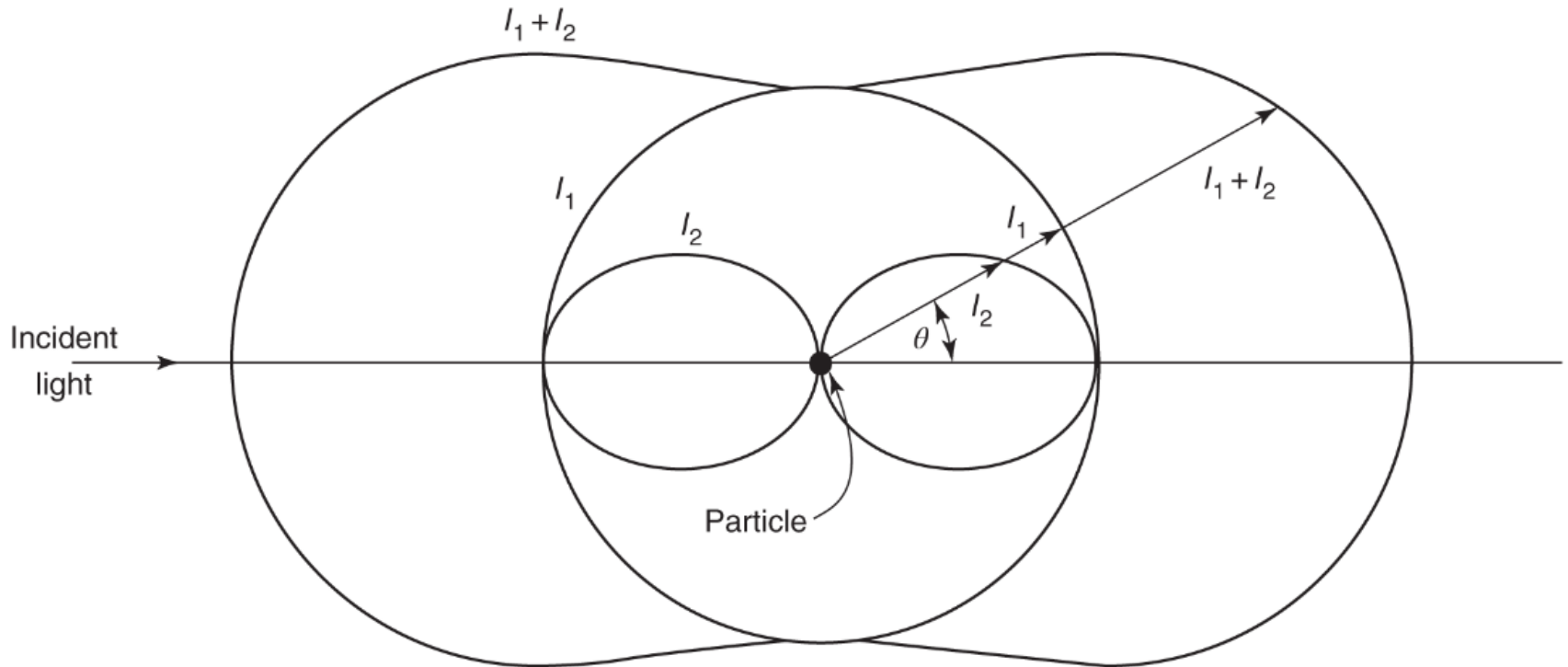
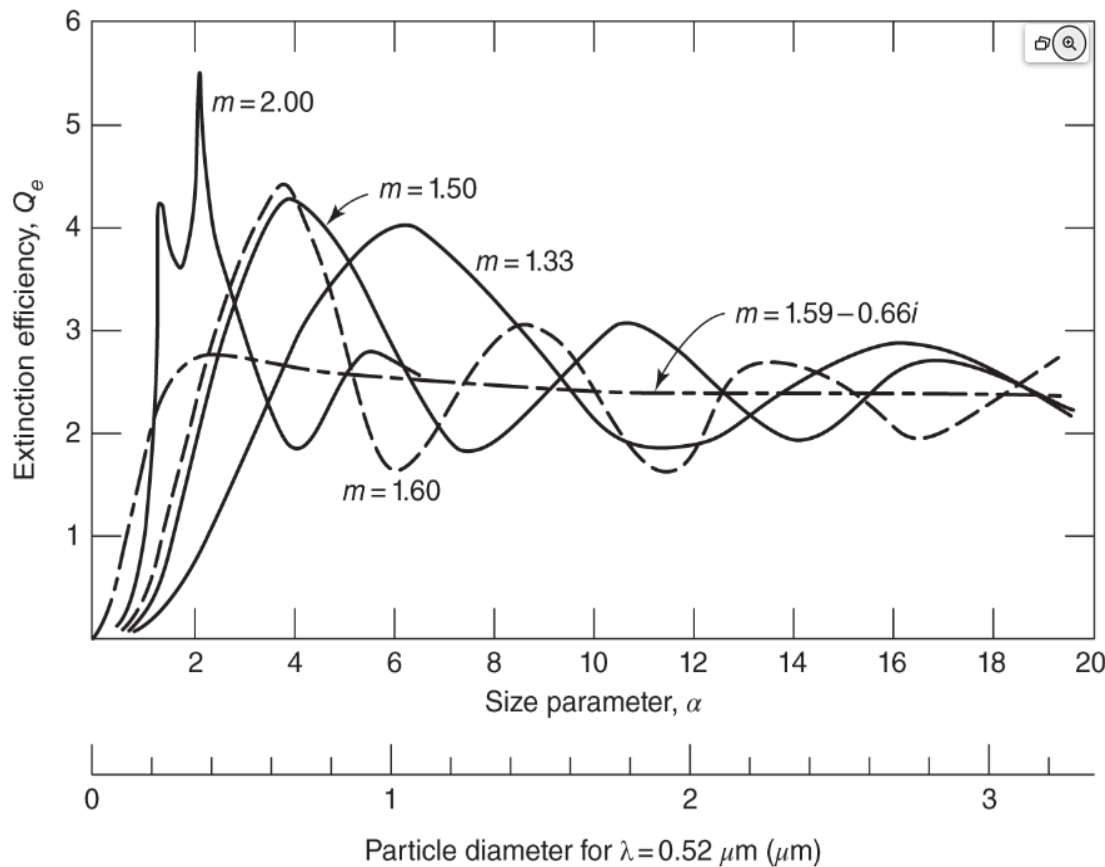


Figure 16.7 Polar diagram of Rayleigh scattered light. Scattering plane is parallel to the paper.

$$I_1 = \frac{I_0 \pi^4 d^6}{8R^2 \lambda^4} \left(\frac{m^2 - 1}{m^2 + 2} \right)^2 \quad \text{for } d < 0.05 \mu\text{m} \quad I_2 = \frac{I_0 \pi^4 d^6}{8R^2 \lambda^4} \left(\frac{m^2 - 1}{m^2 + 2} \right)^2 \cos^2 \theta \quad \text{for } d < 0.05 \mu\text{m}$$

Mie Scattering



[Figure 16.2](#) Extinction efficiency versus particle size for spheres (after Hodkinson, [1966](#)).

$$\alpha = \frac{\pi D}{\lambda} \quad \text{size parameter}$$

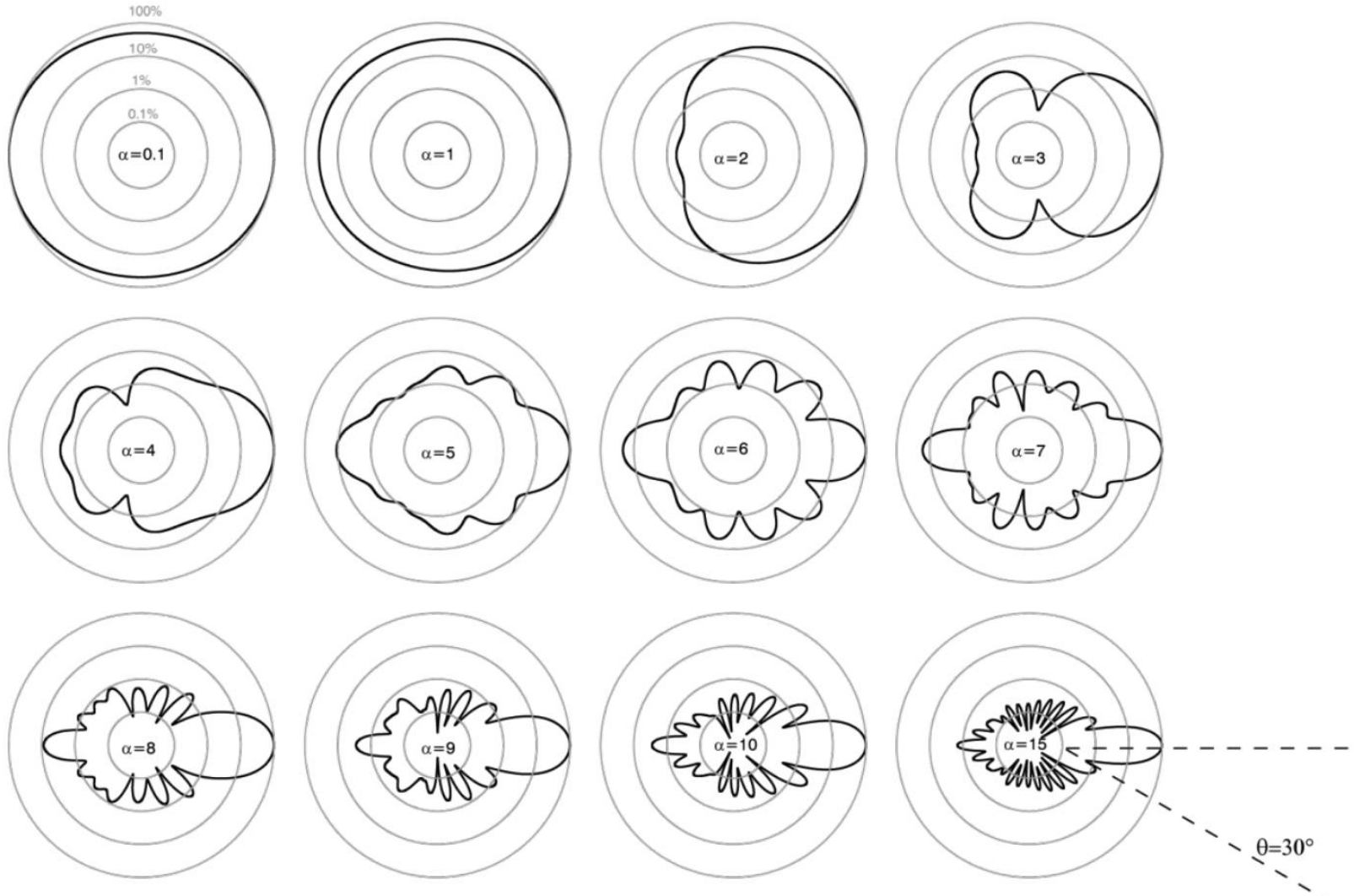
Mie algorithm:

- solves EM field equations
- predicts Q_{scat} , Q_{abs} from α , m
- predicts the phase function for perpendicular and parallel polarization

Main results:

- $\alpha \ll 1 \Rightarrow$ converges to Rayleigh scattering
- $\alpha \gg 1 \Rightarrow$ converges to geometric scattering ($Q_{ext} = 2$)
- $1 < \alpha < 100 \Rightarrow Q_{ext}$ is oscillating and very sensitive to size, wavelength and refractive index

Mie scattering diagram



Particle Absorption

- Particles are made up of different components
 - Each component absorbs differently
 - Characterized by their imaginary index of refraction k
- Particles absorb preferentially at IR wavelengths
 - Cloud effects on surface heating

Particle Absorption

- Only some components of particles absorb visible wavelengths
 - Black/elemental carbon (C)
 - Most important absorber of all visible wavelengths
 - Brown/organic carbon
 - Hematite (Fe_2O_3)
 - In soil dust
 - Aluminum oxide (Al_2O_3)
 - In soil-dust and combustion particles
 - PAHs and nitrated aromatics
 - Absorb blue light preferentially

Ranks

- Clean air:
 - Gas scattering (N_2 , O_2 , Ar, H_2O) most important
 - The other three has similar effects
- Polluted air:
 - Particle scattering (60%-95%)
 - H_2O , SO_4^{2-} , OC, NO_3^-
 - Particle absorption (5%-40%)
 - Black/elemental carbon
 - NO_2 gas absorption (5%-40%)
 - Gas scattering

Total Extinction

- Total Extinction:
 - The attenuation of incident light over a given distance due to absorption and scattering
- Total extinction coefficient
 - Sum of absorption and scattering by gases and particles

$$\sigma_t = \sigma_{a,g} + \sigma_{s,g} + \sigma_{a,p} + \sigma_{s,p}$$

$$C = e^{-\sigma x}$$

Total Extinction

- Total extinction coefficient can be calculated based on PM/gas concentrations (not required)
- Can also be empirically estimated
- A quick example from EPA-454/B-03-005:

$$\begin{aligned}\sigma_t \sim & (3)f(RH)[\text{sulfate}] + (3)f(RH)[\text{nitrate}] \\ & +(4)[\text{Organic carbon mass}] \\ & +(10)[\text{black carbon}] \\ & +(1)[\text{Soil}] + (0.6)[\text{Coarse mass}] + 10\end{aligned}$$

Total Extinction

- Regulatory: unit for σ_t is Mm^{-1}
 - Inverse megameter (10^{-6} m^{-1})
- σ_t can varies from:
 - Clean air: 10 Mm^{-1}
 - Very polluted 1000 Mm^{-1}
- Convert to km^{-1} : $0.01 - 1 \text{ km}^{-1}$
- Deciview also used

$$dv = 10 \ln(\sigma_t/10)$$

Winter and Summer Maps of Light Extinction

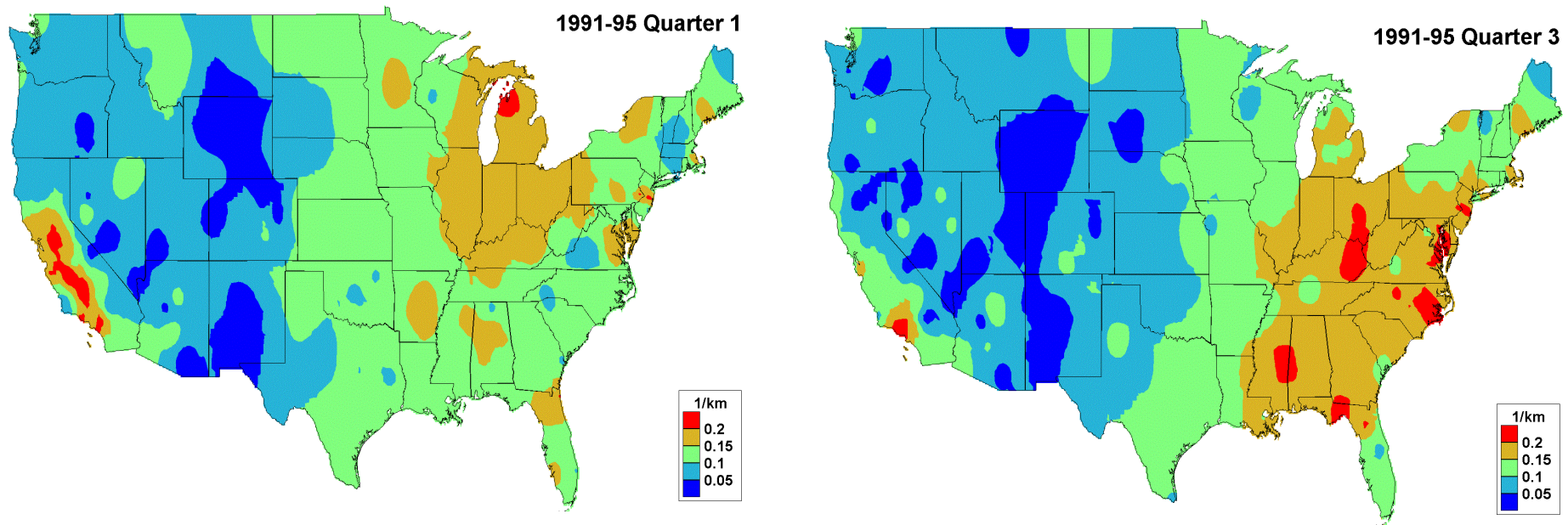


Figure 7.24, Jacobson 2012